

Section : Common

Q.1 The interval on which the function $f(x) = x^3 + 2x^2 - 1$ is decreasing, is

1. $(-\infty, -4/3)$
2. $[0, \infty)$
3. $[-4/3, 0]$
4. $[-4/3, \infty)$

Options 1. 1

2. 2
3. 3
4. 4

Q.2 If A and B are invertible matrices of the same order, then $(AB)^{-1}$ is equal to

1. $A^{-1}B^{-1}$
2. $A^{-1}B$
3. $B^{-1}A^{-1}$
4. AB^{-1}

Options 1. 1

2. 2
3. 3
4. 4

Q.3 If A and B are symmetric matrices of the same order, then which of the following are true?

- (A) $AB - BA$ is a skew symmetric matrix
- (B) AB is a symmetric matrix
- (C) AB is a scalar matrix
- (D) $AB + BA$ is a symmetric matrix

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A) and (D) only
3. (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.4 If $f(x) = \begin{vmatrix} 0 & x-1 & x-2 \\ x+1 & 0 & x-3 \\ x+2 & x+3 & 0 \end{vmatrix}$, then the value of $f(0)$ is equal to :

1. -1
2. 1
3. -2
4. 0

Options 1. 1

2. 2
3. 3
4. 4

Q.5 The particular solution of the differential equation $\frac{dy}{dx} + \frac{3y}{x} = 0, y(1) = 1$ is

1. $y = \frac{1}{x^3}$
2. $y = \frac{1}{x^2}$
3. $x = \frac{1}{y}$
4. $x = \frac{1}{y^2}$

Options 1. 1

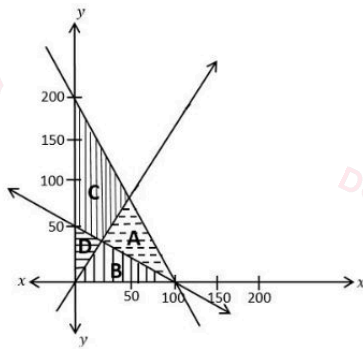
2. 2

3. 3

4. 4

Q.6

The feasible region represented by the constraints: $x + 2y \geq 100$, $2x - y \leq 0$, $2x + y \leq 200$, $x \geq 0$, $y \geq 0$ of an LPP is :



1. Region A
2. Region B
3. Region C
4. Region D

Options 1. 1

2. 2

3. 3

4. 4

Q.7

The value of $\int \frac{x^5}{\sqrt{1+x^3}} dx$ is

1. $\frac{2}{9}(1+x^3)^{3/2} - \frac{2}{3}(1+x^3)^{1/2} + c$: c is an arbitrary constant
2. $\frac{2}{3}(1+x^3)^{3/2} - \frac{2}{3}(1+x^3)^{1/2} + c$: c is an arbitrary constant
3. $\frac{1}{3}(1+x^3)^{3/2} + \frac{1}{3}(1+x^3)^{1/2} + c$: c is an arbitrary constant
4. $\frac{2}{9}(1+x^3)^{3/2} + \frac{2}{3}(1+x^3)^{1/2} + c$: c is an arbitrary constant

Options 1. 1

2. 2

3. 3

4. 4

Q.8

The value of $\int_{-1}^1 |x^3 - x| dx$ is

1. 0
2. $\frac{1}{4}$
3. $\frac{1}{2}$
4. 1

Options 1. 1

2. 2

3. 3

4. 4

Q.9

Which one of the following equations is a homogeneous differential equation?

1. $(4x + 5)dy + (3y - 4)dx = 0$
2. $x^2y dx - (x^3 + y^3) dy = 0$
3. $(x^3 + 2y^2x) dy + 2xy dx = 0$
4. $x^2 dy - y^2 dx = \sqrt{x^2 + y^2} dx$

Options 1. 1

2. 2

3. 3

4. 4

Q.10

Two cards are drawn simultaneously at random from a well shuffled pack of 52 Cards.

Let X be the random variable which denotes number of kings in the draw. Then the probability distribution of X is

(1)

X	0	1	2
P(X)	$\frac{{}^{48}C_2}{{}^{52}C_2}$	$\frac{{}^4C_1 \times {}^{48}C_1}{{}^{52}C_2}$	$\frac{{}^4C_2}{{}^{52}C_2}$

(2)

X	0	1	2
P(X)	$\frac{{}^4C_2}{{}^{52}C_2}$	$\frac{{}^4C_1 \times {}^{48}C_1}{{}^{52}C_2}$	$\frac{{}^4C_2}{{}^{52}C_2}$

(3)

X	0	1	2
P(X)	$\frac{{}^4C_2}{{}^{52}C_2}$	$\frac{{}^4C_1 \times {}^{48}C_1}{{}^{52}C_2}$	$\frac{{}^{48}C_2}{{}^{52}C_2}$

(4)

X	0	1	2
P(X)	$\frac{{}^4C_0}{{}^{52}C_2}$	$\frac{{}^{48}C_1}{{}^{52}C_2}$	$\frac{{}^4C_2}{{}^{52}C_2}$

- Options
- 1
 - 2
 - 3
 - 4

Q.11 If $y = (x+1)(x^2+1)(x^4+1)(x^8+1)$ then $\frac{dy}{dx}$ at $x = -1$ is

- 8
- 8
- 16
- 16

- Options
- 1
 - 2
 - 3
 - 4

Q.12 Let $A = [a_{ij}]$ is given by $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 4 & -5 \\ 2 & -1 & 3 \end{bmatrix}$. Then the matrix $B = [b_{ij}]$, where b_{ij} = Minor of a_{ij} is:

- $\begin{bmatrix} 7 & 19 & -11 \\ -1 & -1 & 1 \\ -3 & -11 & 7 \end{bmatrix}$
- $\begin{bmatrix} 7 & -19 & 11 \\ 5 & -1 & -1 \\ 2 & 11 & 7 \end{bmatrix}$
- $\begin{bmatrix} 7 & -5 & -3 \\ 19 & 1 & -11 \\ -11 & 7 & 7 \end{bmatrix}$
- $\begin{bmatrix} 7 & 19 & -11 \\ -3 & 11 & 7 \\ -2 & -1 & -1 \end{bmatrix}$

- Options
- 1
 - 2
 - 3
 - 4

Q.13 The area of the region bounded by the parabola $y^2 = x$ and the straight line $2y = x$ is

- $\frac{7}{3}$ sq. units
- $\frac{5}{3}$ sq. units

3. $4/3$ sq. units

4. $2/3$ sq. units

Options 1. 1

2. 2

3. 3

4. 4

Q.14 The corner points of the bounded feasible region determined by the system of linear constraints are $(15, 0)$, $(40, 0)$, $(4, 18)$ and $(6, 12)$. If objective function is $Z = 30x + 20y$, then the sum of the maximum and the minimum values of Z is

1. 900

2. 1650

3. 1620

4. 1680

Options 1. 1

2. 2

3. 3

4. 4

Q.15 For the function $f(x) = e^{-2x}(2 - x)^2$, the point of local maxima is:

1. $x = 1$

2. $x = 2$

3. $x = 3$

4. $x = 5/2$

Options 1. 1

2. 2

3. 3

4. 4

Section : Applied Mathematics

Q.16 Match List-I with List-II

List I

(A) Corner point of a feasible region

(B) Bounded feasible region

(C) Unbounded feasible region

(D) Convex region

List II

(I) The line segment joining any two arbitrary points of the region always lies entirely within the region

(II) can not be enclosed within a circle

(III) can be enclosed within a circle

(IV) is a point of intersection of two boundary lines in the feasible region

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)

2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

4. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.17 Two pipes can fill a tank in 6 minutes and 12 minutes respectively, and a third pipe can empty the tank at the rate of 18 liters per minute. If all the pipes working together can fill the empty tank in 5 minutes, the capacity of the tank is:

1. 600 litres

2. 480 litres

3. 360 litres

4. 630 litres

Options 1. 1

2. 2

3. 3

4. 4

Q.18 A cylindrical drum of radius 7 cm and height 2 m is being kept in a vertical position filled with milk. If the milk is leaking at $14 \text{ cm}^3/\text{sec}$ from its lower base, then the rate of decrease in the level of milk is:

[Take $\pi = \frac{22}{7}$]

1. $\frac{2}{11}$ cm/sec
2. $\frac{1}{11}$ cm/sec
3. $\frac{3}{11}$ cm/sec
4. $\frac{4}{11}$ cm/sec decreases

Options 1. 1

2. 2
3. 3
4. 4

Q.19

If matrix $A = \begin{bmatrix} x & 2 & 3 \\ a & y & -5 \\ b & c & 0 \end{bmatrix}$ is a skew-symmetric matrix, then

- (A) $x + y + c = 5$
- (B) $c = 5$
- (C) $a + b + c = 0$
- (D) $a + b - c = 10$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A), (B) and (C) only
3. (D) and (A) only
4. (A) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.20 Let A be a non-singular square matrix of order n , then

List I

List II

(A) $A(\text{adj } A)$

(I) $\frac{1}{|A|}$

(B) $|\text{adj } A|$

(II) $|A|^n$

(C) $|A^{-1}|$

(III) $|A|I$

(D) $|A(\text{adj } A)|$ (IV) $|A|^{n-1}$

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.21 Consider the following hypothesis test.

$H_0 : \mu \leq 12$

$H_a : \mu > 12$

If a sample of 25 is taken with sample mean 15 and a sample standard deviation of 6, then the value of t-test statistic is:

1. $\frac{3}{4}$
2. $\frac{1}{2}$
3. $\frac{3}{2}$

4. $\frac{5}{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.22 For the system of linear equations

$$x + y + z = 5000$$

$$6x + 7y + 8z = 35800$$

$$6x + 7y - 8z = 7000$$

the values of x, y and z are:

1. $x = 1800; y = 1000; z = 2200$

2. $x = 1000; y = 1800; z = 2200$

3. $x = 1000; y = 2200; z = 1800$

4. $x = 2200; y = 1800; z = 1000$

Options 1. 1

2. 2

3. 3

4. 4

Q.23 If A is a square matrix such that $A^2 = A$ and I is the identity matrix of the same order as A , then $(I + A)^2 - 3A$ is equal to

1. I

2. $2A$

3. $3I$

4. A

Options 1. 1

2. 2

3. 3

4. 4

Q.24 Mr. Jayesh plans to save amount for higher studies of his daughter, required after 10 years. How much amount should he save at the beginning of each year to accumulate Rs.1,00,000 at the end of 10 years, if rate of interest is 12% compounded annually?

[Given $(1.12)^{11} = 3.5$]

1. Rs. 5291

2. Rs. 5000

3. Rs. 5042

4. Rs. 5500

Options 1. 1

2. 2

3. 3

4. 4

Q.25 Rs. 2,50,000 cash is equivalent to a perpetuity of Rs. 7500 payable at the end of each quarter. Then the rate of interest is

1. 4% p.a

2. 6% p.a

3. 3% p.a

4. 12% p.a

Options 1. 1

2. 2

3. 3

4. 4

Q.26 The five month moving averages for the following data

Month (rth) 43 44 45 46 47 48 49 50 51

Actual Demand 105 106 110 110 114 121 130 128 137

is:

1. 109, 112.2, 117, 120.6, 126

2. 109, 112, 117, 123, 128

3. 109, 112, 119, 126, 135

4. 109, 112.2, 117, 120, 131

Options

1. 1
2. 2
3. 3
4. 4

Q.27 Consider the following hypothesis test:

$$H_0 : \mu \leq 3432$$

$$H_a : \mu > 3432$$

A sample of 96 provided a sample mean $\bar{x} = 3648$ and sample standard deviation $s = 802$ then the degree of freedom of t-distribution is:

1. 95
2. 97
3. 48
4. 47

Options

1. 1
2. 2
3. 3
4. 4

Q.28 For the objective function $Z = 3x + 5y$ subject to constraints $x + 3y \geq 3$, $x + y \geq 2$, $x \geq 0$, $y \geq 0$:

1. there exists only maximum value of Z
2. there exists only minimum value of Z
3. there exists both maximum and minimum values of Z
4. there is neither maximum value nor minimum value of Z

Options

1. 1
2. 2
3. 3
4. 4

Q.29 A swimmer whose speed in swimming pool is 5 km/h, swims between two points in a river and returns back to starting point. He took 20 minutes more to cover the distance upstream than to cover downstream. If the speed of stream is 2 km/h, then the distance between two points is

1. 1.75 km
2. 0.233 km
3. 35 km
4. 0.25 km

Options

1. 1
2. 2
3. 3
4. 4

Q.30 If the area above x-axis, bounded by the curves $y = 3^{\beta x}$, $x = 0$ and $x = \frac{26}{\log_e 3}$, then the value of β is:

1. $\beta = 2$
2. $\beta = 1$
3. $\beta = -1$
4. $\beta = \frac{1}{2}$

Options

1. 1
2. 2
3. 3
4. 4

Q.31 Let X be a random variable. Let $E(X)$ and $\text{Var}(X)$ denote the mean and the variance of X respectively.

List I

(A) If $\text{Var}(X) = \alpha$, then $\text{Var}(2X + 3)$ is

(B) If $E(X) = \alpha$, then $E(2X)$ is

(C) If $\text{Var}(X) = \alpha$, then $\text{Var}(3X - \alpha) + \text{Var}(\sqrt{2}X + \beta)$ is

List II

(I) 11α

(II) 6α

(III) 4α

If $E(X) = \frac{5\alpha}{12}$, then $E(12X + \alpha)$ is (IV) 2α

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.32 For independent events $A_1, A_2, A_3, \dots, A_n$, if $P(A_i) = \frac{1}{i+1}$, $i = 1, 2, 3, \dots, n$, then the probability that none of the events occur is:

1. $\frac{n}{n+1}$
2. $\frac{n-1}{n+1}$
3. $\frac{1}{n+1}$
4. $\frac{n-1}{2(n+1)}$

Options 1. 1

2. 2
3. 3
4. 4

Q.33 The largest open interval in which the function $f(x) = 4x^3 - 5x^2 - 8x + 12$ increases, is:

1. $(-\infty, -\frac{4}{3}) \cup (\frac{4}{3}, \infty)$
2. $(-\infty, \frac{1}{2}) \cup (\frac{4}{3}, \infty)$
3. $(-\infty, \frac{1}{2}) \cap (\frac{4}{3}, \infty)$
4. $(-\infty, \frac{1}{2}) \cup (\frac{4}{3}, \infty)$

Options 1. 1

2. 2
3. 3
4. 4

Q.34 A person has set up a sinking fund in order to have Rs. 10,00,000 after 10 years for his child education. The amount should put bi-annually into account paying 5% per annum compounded semi-annually is: [Given $(1.025)^{20} = 1.6386$]

1. Rs. 29148.14
2. Rs. 39148.14
3. Rs. 35148.14
4. Rs. 36148.14

Options 1. 1

2. 2
3. 3
4. 4

Q.35 In a survey question for a sample of 250 individuals, 120 persons gave response 'yes', 80 persons gave response 'no' and 50 gave 'no response'. The point estimate of the proportion in the population who responded 'yes' is:

1. 0.32
2. 0.2
3. 0.48
4. 0.8

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Mohini purchases a house worth Rs. 50 lakhs and makes a down payment of Rs. 11.2 lakhs. She pays the remaining amount on monthly EMI using a reducing balance method. The bank charges 6\% per annum compounded monthly for a tenure of 25 years. Her EMI is:
[Given: $(1.005)^{-300} \approx 0.224$]

1. Rs. 11200
2. Rs. 38800
3. Rs. 25000
4. Rs. 22400

Options 1. 1
2. 2
3. 3
4. 4

Q.37 It is given that 3\% of items manufactured by an industry are defective. The probability that a packet of 250 items contains one defective item is:
[Given: $e^{-7.5} = 0.000553$]

1. 0.00412
2. 0.00312
3. 0.00415
4. 0.00315

Options 1. 1
2. 2
3. 3
4. 4

Q.38 The water and milk in two vessels are in the ratio 1 : 1 and 3 : 8 respectively. In what ratio, the mixtures in the vessels be mixed to obtain a new mixture containing water and milk in the ratio 4 : 7?

1. 1 : 4
2. 2 : 3
3. 3 : 4
4. 4 : 1

Options 1. 1
2. 2
3. 3
4. 4

Q.39 If an asset costs Rs. 50,000 with an estimated useful life of 6 years and a scrap value of Rs. 5000. Then by using a linear depreciation method, the annual depreciation of the asset will be :

1. Rs. 7,000
2. Rs. 7,500
3. Rs. 7,800
4. Rs. 8,000

Options 1. 1
2. 2
3. 3
4. 4

Q.40 Ram had invested Rs. 15,000 in a mutual fund and the value of the investment at the time of redemption was Rs. 25,000. If the compound annual growth rate (CAGR) is 8.88\%, then the number of years for which Ram has invested the amount is:
[Given: $\log 1.089 = 0.0370$ and $\log 1.667 = 0.2220$]

1. 6 years
2. 5 years
3. 7 years
4. 4 years

Options 1. 1
2. 2
3. 3
4. 4

Q.41 A fair coin is tossed a fixed number of times. If the probability of getting 11 heads is equal to the probability of getting 13 heads, then the probability of getting 2 heads is:

1. $\frac{69}{2^{22}}$
2. $\frac{23}{2^{24}}$
3. $\frac{69}{2^{24}}$
4. $\frac{23}{2^{22}}$

Options 1. 1

2. 2
3. 3
4. 4

Q.42 A runs 9 times slower than B. If B gives A a start of 80 meters, how far must be the winning post on the tracks so that A and B reach there at the same time?

1. 80 meters
2. 9 meters
3. 98 meters
4. 90 meters

Options 1. 1

2. 2
3. 3
4. 4

Q.43 Match List-I with List-II

List I

(Example)

(A) Lockouts and strikes

(B) Rise and fall of share market

(C) Continuous decline in death rate

(D) The rise in prices before Diwali

List II

(Component of Time Series)

(I) Secular trend

(II) Seasonal trend

(III) Irregular trend

(IV) Cyclical trend

Choose the correct answer from the options given below:

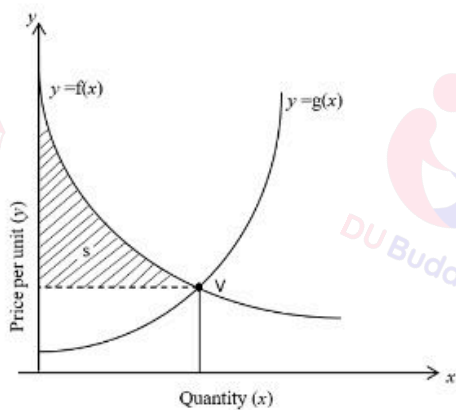
1. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
2. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
3. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
4. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.44

As per the graph given below:



Match **List-I** with **List-II**

List-I	List-II
(Function/Area/point)	(Representation)
(A) Consumers Surplus	(I) $y=g(x)$
(B) Supply function	(II) v
(C) Demand function	(III) s
(D) Equilibrium point	(IV) $y=f(x)$

Choose the **correct** answer from the options given below:

- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.45

The marginal cost of production of x units of a commodity is $56 + \frac{3}{2}x$.

It is known that fixed costs are Rs.115.

Then the total cost of producing 50 units is:-

- Rs. 2800
- Rs. 1875
- Rs. 4790
- Rs. 3383.75

Options 1. 1

2. 2

3. 3

4. 4

Q.46 Consider the following data

Year(X)	2014	2016	2018	2020	2022	2024
Profit(in Rs. Thousand) (Y)	7	9	10	12	14	14

Then for the above data the equation of straight line trend by method of least square is given by:

- $17y = 13x + 285$
- $13y = 17x + 285$

3. $35y = 26(x - 2019) + 385$

4. $26y = 35(x - 2019) + 385$

Options 1. 1

2. 2

3. 3

4. 4

Q.47

If $\begin{vmatrix} -1 & 1 & 0 \\ a & b & 1 \\ 1 & 2 & 1 \end{vmatrix}$ is a singular matrix, then the relation between a and b is:

1. $2a = b$

2. $a + b = 0$

3. $a + b = 3$

4. $a + b = ab$

Options 1. 1

2. 2

3. 3

4. 4

Q.48

If $525 \equiv (10 + K) \pmod{7}$ where $K \in \mathbb{N}$, then the least value of K is:

1.

2. 4

3. 5

4. 6

Options 1. 1

2. 2

3. 3

4. 4

Q.49

The value of $\int_{-5}^5 |x + 3| dx$ is

1. 32

2. 36

3. 34

4. 50

Options 1. 1

2. 2

3. 3

4. 4

Q.50

If $x, y \in \mathbb{R}$ then match List-I with List-II

List I List II

(A) $|x| < |y|$ (I) iff $x^2 > y^2$

(B) $|x| > |y|$ (II) iff $x^2 \leq y^2$

(C) $|x| \leq |y|$ (III) iff $x^2 < y^2$

(D) $|x| \geq |y|$ (IV) iff $x^2 \geq y^2$

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

2. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

4. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Section : Core Mathematics

Q.51

If $f(x) = \sin x - \cos x, x \in [0, 2\pi]$ then

(A) $f(x)$ is increasing in $\left(0, \frac{3\pi}{4}\right)$

(B) $f(x)$ is decreasing in $\left(0, \frac{3\pi}{4}\right)$

(C) $f(x)$ is decreasing in $\left(\frac{3\pi}{4}, \frac{7\pi}{4}\right)$

(D) $f(x)$ is decreasing in $\left(\frac{7\pi}{4}, 2\pi\right)$

Choose the correct answer from the options given below:

1. (A), (D) and (C) only
2. (B), (C) and (D) only
3. (A) and (C) only
4. (B) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.52 The area (in sq.units) of the region enclosed by the curve $y = \cos x$, $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ and the x - axis is:

1. 1

2. 2

3. 4

4. 3

Options 1. 1

2. 2

3. 3

4. 4

Q.53 The area (in sq.units) of the region bounded by the curves $y = x$ and $y = x^3$ is:

1. 1 sq. units

2. $\frac{3}{4}$ sq. units

3. $\frac{1}{2}$ sq. units

4. $\frac{1}{4}$ sq. units

Options 1. 1

2. 2

3. 3

4. 4

Q.54 $\int e^{-x}(\cot x + \operatorname{cosec}^2 x) dx =$

1. $e^{-x} \cot x + c$, where c is an arbitrary constant.

2. $-e^{-x} \cot x + c$, where c is an arbitrary constant.

3. $-e^{-x} \csc^2 x + c$, where c is an arbitrary constant.

4. $e^{-x} \csc^2 x + c$, where c is an arbitrary constant.

Options 1. 1

2. 2

3. 3

4. 4

Q.55 If $\mathbf{A}$ and $\mathbf{B}$ square matrices of order 3 such that $|\mathbf{A}| = -1$, $|\mathbf{B}| = 5$, then the value of $|2\mathbf{AB}|$ is:

1. - 40
2. - 10
3. 20
4. - 20

Options

1. 1
2. 2
3. 3
4. 4

Q.56 Which of the following statements are true?

(A) The vector joining the points $P(2, 3, 0)$ and $Q(-1, -2, -4)$ directed from P to Q is $\overrightarrow{PQ} = -3\mathbf{i} - 5\mathbf{j} - 4\mathbf{k}$

(B) Projection of a vector $\vec{a}$ on other vector $\vec{b}$ is $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$

(C) If $\vec{a} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\vec{b} = -2\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$ then $\vec{a} + \vec{b} = -\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$

(D) If θ is the angle between $\vec{a}$ and $\vec{b}$ then $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (C), (B) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.57 Match List-I with List-II

List-I

(A) If vectors $\vec{a}$ and $\vec{b}$ are such that $\vec{a} = \lambda\vec{b}$ and $|\vec{a}| = |\vec{b}|$, then

(B) Projection vector of $\vec{a}$ on $\vec{b}$

(C) If $\vec{a}$ and $\vec{b}$ are non-zero vectors such that $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, then

(D) If $|\vec{a}| = 4$ and $-3 \leq \lambda \leq 2$, then the range of $|\lambda\vec{a}|$

List-II

(I) $\vec{a}$ and $\vec{b}$ are orthogonal

(II) $[0, 12]$

(III) $\vec{a} = \pm\vec{b}$

(IV) $\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}\right)\vec{b}$

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
2. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options

1. 1
2. 2
3. 3
4. 4

Q.58 If $x = a \sec^3 \theta$, $y = a \tan^3 \theta$, then $\frac{d^2y}{dx^2}$ equals.

1. $\frac{\cos^5 \theta}{3a \sin \theta}$

2. $\frac{3a \cos^5 \theta}{\sin \theta}$
3. $\frac{\sin^5 \theta}{3a \cos \theta}$
4. $\frac{3a \sin^5 \theta}{\cos \theta}$

Options 1. 1
2. 2
3. 3
4. 4

Q.59 Match List-I with List-II

List-I

List-II

(A)

(I)

Line: $x = 2y + 1 = z - 1$ Crosses xz -plane at $(1, 0, 1)$

(B)

(II)

Line: $x + 1 = 2y + 1 = z$ Crosses xz -plane at $(0, 0, 1)$

(C)

(III)

Line: $x - 1 = 2y = z + 1$ Crosses xz -plane at $(1, 0, -1)$

(D)

(IV)

Line: $x - 1 = 2y = z - 1$ Crosses xz -plane at $(1, 0, 2)$

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
2. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
4. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

Options 1. 1
2. 2
3. 3
4. 4

Q.60

If c_{ij} denotes the cofactor of element a_{ij} of the matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 2 \\ 4 & 2 & 3 \end{bmatrix}$, then the value of $c_{21} \cdot c_{33}$ is

1. -8
2. -3
3. -24
4. 24

Options 1. 1
2. 2
3. 3
4. 4

Q.61 A coin is tossed and a die is thrown. The probability that the outcome will be a tail on the coin or a number greater than 3 on the die is

1. $\frac{3}{4}$
2. $\frac{1}{4}$
3. $\frac{1}{2}$
4. 0

Options 1. 1
2. 2
3. 3
4. 4

Q.62 For $x \in \mathbb{R}$, if $f(x) = -(x - 1)^2 + 2$, then

- (A) f is an increasing function on $(-\infty, 1]$
- (B) f has no critical points
- (C) f has a maximum value at $x = 1$

(D) f has a minimum value at $x = 1$

Choose the correct answer from the options given below:

1. (B) and (D) only
2. (B), (C) and (D) only
3. (A) and (C) only
4. (A) and (B) only

Options

1. 1
2. 2
3. 3
4. 4

Q.63 Match List-I with List-II

List-I

List-II

- (A) The number of possible matrices of order 3×3 with each entry 1 or 0 (I) 2^4
- (B) The number of possible matrices of order 2×3 with each entry 1 or 0 (II) 2^9
- (C) The number of possible matrices of order 2×3 with each entry 0, 1, 2 (III) 2^6
- (D) The number of possible matrices of order 2×2 with each entry 1 or 0 (IV) 3^6

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
2. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
3. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
4. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

Options

1. 1
2. 2
3. 3
4. 4

Q.64 If $\begin{vmatrix} p-a & 0 & c-r \\ 0 & q-b & c-r \\ a & b & r \end{vmatrix} = 0$ then the value of $\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c}$ is

1. 0
2. 1
3. 2
4. 3

Options

1. 1
2. 2
3. 3
4. 4

Q.65 If A and B are independent events, then which of the following is/are true?

- (A) $\bar{A}$ and B are independent events
- (B) $P(A \cap B) = 0$
- (C) $\bar{A}$ and $\bar{B}$ are independent events
- (D) $P(A \cap B) = P(A) + P(B)$

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (A), (B) and (D) only
3. (B) and (D) only
4. (C) only

Options

1. 1
2. 2
3. 3
4. 4

Q.66 The projection of the vector $5\hat{i} + \hat{j} - 3\hat{k}$ on the vector $\hat{i} + 2\hat{j} - 3\hat{k}$ is

1. $\frac{16}{\sqrt{14}}$

2. $\frac{8}{7}$
3. $\frac{15}{13}$
4. $\frac{16}{\sqrt{35}}$

Options 1. 1

2. 2
3. 3
4. 4

Q.67 If $\vec{a}$ is a unit vector and $(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 15$, then the value of $|\vec{x}|$ is:

1. 0
2. 16
3. 4
4. 2

Options 1. 1

2. 2
3. 3
4. 4

Q.68 A balloon which always remains spherical, has a variable diameter, $\frac{3}{2}(5x + 7)$. Then the rate of change of its volume with respect to x is

1. $\frac{27}{8}(5x + 7)^2$
2. $\frac{27}{16}(5x + 7)^2$
3. $\frac{135\pi}{8}(5x + 7)^2$
4. $\frac{135\pi}{16}(5x + 7)^2$

Options 1. 1

2. 2
3. 3
4. 4

Q.69 The corner points of the bounded feasible region determined by a set of constraints in an LPP are $P(0, 5)$, $Q(3, 5)$, $R(5, 0)$ and $S(4, 1)$.

If the objective function is $z = ax + 2by$, where $a, b > 0$, then the condition on a and b such that the maximum value of z occurs at Q and S is

1. $a - 5b = 0$
2. $a - 3b = 0$
3. $a - 2b = 0$
4. $a - 8b = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.70 If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \cdots \infty}}}$ then $\frac{dy}{dx}$ equals to

1. $\frac{-\cos x}{2y - 1}$
2. $\frac{\cos x}{2y - 1}$
3. $\frac{\sin x}{1 - 2y}$
4. $\frac{1 - 2y}{\cos x}$

Options 1. 1

2. 2
3. 3
4. 4

Q.71 The function $f : \mathbb{R} \rightarrow [-1, 1]$ defined by $f(x) = \cos x$ is:

1. one-one but not onto
2. onto but not one-one
3. Both one-one and onto
4. Neither one-one nor onto

Options 1. 1

2. 2
3. 3
4. 4

Q.72 Let L_1 and L_2 be two lines, represented as,

$L_1 : \vec{r} = \hat{i} + \hat{j} + \lambda(2\hat{i} - \hat{j} + \hat{k})$ and $L_2 : \vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu(4\hat{i} - 2\hat{j} + 2\hat{k})$, where λ and μ are scalars. Then which of the following are true?

- (A) L_1 is perpendicular to L_2
- (B) L_1 is parallel to L_2
- (C) L_1 passes through the point $(1, 1, 0)$
- (D) L_2 passes through the point $(2, 1, -1)$

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (B), (C) and (D) only
3. (C) and (D) only
4. (A) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.73 Match List-I with List-II

List-I

List-II

(A) The degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right) = x \sin\left(\frac{dy}{dx}\right)$ (I) 4

(B) The degree of differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^{1/4} + x^{1/5} = 0$ (II) 1

(C) The degree of differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 + 6y^5 = 0$ (III) Not defined

(D) The degree of differential equation $1 + \left(\frac{dy}{dx}\right)^4 = 7\left(\frac{d^2y}{dx^2}\right)^3$ (IV) 3

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.74 If $P(A) = \frac{3}{5}$, $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{4}$, then $P(\bar{A} | \bar{B})$ is

1. $\frac{3}{40}$
2. $\frac{3}{10}$
3. $\frac{17}{20}$
4. $\frac{17}{40}$

Options 1. 1

2. 2
3. 3
4. 4

Q.75 The following system of equations:

$$x + y - z = 7$$

$$4x + \lambda y - \lambda z = 3$$

$$3x + 2y - 4z = 5$$

does not possess a solution if the value of λ is:

1. 3
2. -3
3. -4
4. 4

Options 1. 1

2. 2
3. 3
4. 4

Q.76 If $f(x) = \begin{cases} ax - 1 & \text{if } x \geq 1 \\ 2x + 1 & \text{if } x < 1 \end{cases}$ is continuous at $x = 1$, then a equals

1. 0
2. 3
3. 4
4. 2

Options 1. 1

2. 2
3. 3
4. 4

Q.77 $\int \frac{e^{2x} - 1}{e^{2x} + 1} dx =$

1. $\log|e^x + e^{-x}| + C$: C is an arbitrary constant
2. $\log|e^{2x} + 1| + C$: C is an arbitrary constant
3. $\log|e^{2x} - e^{-x}| + C$: C is an arbitrary constant
4. $\log|e^{2x} - 1| + C$: C is an arbitrary constant

Options 1. 1

2. 2
3. 3
4. 4

Q.78 If A is a square matrix such that $A^2 = A$ and I is the identity matrix of same order as A , then the value of

$(A - 2I)^2 - (2A + I)^2 + 11A$ is:

1. I
2. $2I$
3. $3I$
4. $-I$

Options 1. 1

2. 2
3. 3
4. 4

Q.79 Match List-I with List-II

List-I

List-II

(A) $\tan^{-1}\left(\frac{2}{11}\right) + \tan^{-1}\left(\frac{7}{24}\right)$ (I) $\frac{3\pi}{4}$

(B) $\tan^{-1}2 + \tan^{-1}3$ (II) π

(C) $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3$ (III) $\tan^{-1}\left(\frac{1}{2}\right)$

(D) $\tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right)$ (IV) $\tan^{-1}\left(\frac{2}{9}\right)$

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
3. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
4. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.80 Cosine of the acute angle between the lines $\frac{x-3}{2} = \frac{y-2}{1} = \frac{z-5}{2}$ and $\frac{x-1}{6} = \frac{y-3}{-3} = \frac{z+6}{2}$ is

1. $\frac{11}{21}$
2. $\frac{13}{21}$
3. $\frac{10}{21}$
4. $\frac{8}{21}$

Options 1. 1

2. 2
3. 3
4. 4

Q.81 $\int \frac{dx}{e^x + e^{-x}}$ is equal to

1. $\tan^{-1}(e^x) + c$: c is an arbitrary constant
2. $\tan^{-1}(e^{-x}) + c$: c is an arbitrary constant
3. $\log(e^x - e^{-x}) + c$: c is an arbitrary constant
4. $\log(e^x + e^{-x}) + c$: c is an arbitrary constant

Options 1. 1

2. 2

3. 3

4. 4

Q.82 Let box I contains 3 black and 4 white balls, box II contains 2 black and 2 white balls, box III contains 4 black and 3 white balls. A box is selected at random and then a ball is randomly drawn from the selected box. If the color of the ball is black, then the probability that the ball is drawn from box III, is:

1. $\frac{1}{7}$
2. $\frac{4}{21}$
3. $\frac{8}{21}$
4. $\frac{9}{21}$

Options 1. 1

2. 2

3. 3

4. 4

Q.83 The solution of the differential equation $(x^2 + xy)dy = (x^2 + y^2)dx$ is

1. $y + \log_e \left| \frac{y-2}{x} \right| = c$, (c is an arbitrary constant)
2. $\frac{y}{x} + \log_e \left| \frac{(y-x)^2}{x} \right| = c$, (c is an arbitrary constant)
3. $y - \log_e \left| \frac{y-x}{x} \right| = c$, (c is an arbitrary constant)
4. $y^2 + \log_e \left| \frac{(y-x)^2}{x} \right| = c$, (c is an arbitrary constant)

Options 1. 1

2. 2

3. 3

4. 4

Q.84 For the LPP: minimize $z = 6x + 3y$ subject to the constraints

$$4x + y \geq 80$$

$$x + 5y \geq 115$$

$$3x + 2y \leq 150$$

$x \geq 0, y \geq 0$ then the minimum value of z is

1. 150
2. 228
3. 120
4. 100

Options 1. 1

2. 2

3. 3

4. 4

Q.85 The domain of the function $y = \sin^{-1}(x-1) + \cos^{-1}\sqrt{x-1}$ is:

1. $[-1, 1]$
2. $[0, 1]$
3. $[1, 2]$
4. $[0, 2]$

Options 1. 1

2. 2

3. 3

4. 4